

RA8P1 E-READER

Processor schematic development
Power networks are being implemented from RA8P1-specific references.
This revision is not a completed schematic or a fabrication release.

RA8P1 IO allocation



File: mcu_interfaces.kicad_sch

Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

Remaining subsystem circuits follow epic #821 and issues #824-#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

Footprint and PCB qualification are deferred by scope.

RA8P1 internal core regulator



File: mcu_core_power.kicad_sch

RA8P1 IO and analog supplies



File: mcu_supply_power.kicad_sch

RA8P1 clocks, reset and debug



File: mcu_clocks_debug.kicad_sch

Sheet: / File: ereader_rev1.kicad_sch		
Title: RA8P1 e-reader - schematic index		
Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.5	Id: 1/5	

DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.

U1B
R7KA8P1KFLCAC#UC0

GPIO P0 / P1		
U13	P000	P100
T13	P001	P101
U14	P002	P102
R12	P003	P103
T11	P004	P104
R11	P005	P105
U12	P006	P106
T12	P007	P107
U11	P008	P108
P11	P009	P109
P10	P010	P110
N10	P011	P111
P9	P012	P112
N9	P013	P113
T8	P014	P114
U8	P015	P115

U1C
R7KA8P1KFLCAC#UC0

GPIO P2 / P3		
C5	P200	P300
B15	P206	P301
A17	P207	P302
		P303
		P304
		P305
		P306
		P307
		P308
		P309
		P310
		P311
		P312

U1D
R7KA8P1KFLCAC#UC0

GPIO P4 / P5		
P17	P400	P500
N17	P401	P501
L14	P402	P502
H13	P403	P503
J13	P404	P504
G12	P405	P505
F14	P406	P506
R17	P407	P507
M15	P408	P508
R16	P409	P509
K13	P410	P510
M14	P411	P511
M12	P412	P512
P14	P413	P513
L13	P414	P514
R15	P415	P515

U1E
R7KA8P1KFLCAC#UC0

GPIO P6 / P7		
P3	P600	P700
P4	P601	P701
P2	P602	P702
P1	P603	P703
N2	P604	P704
N1	P605	P705
N3	P606	P706
M2	P607	P707
K2	P608	P708
A1	P609	P709
D3	P610	P710
D2	P611	P711
E4	P612	P712
C2	P613	P713
E3	P614	P714
E2	P615	P715

U1F
R7KA8P1KFLCAC#UC0

GPIO P8 / P9		
T6	P800	P902
P6	P801	P903
R6	P802	P904
P7	P803	P905
R7	P804	P906
P12	P805	P907
T14	P806	P908
N11	P807	P909
U5	P808	P910
T5	P809	P911
M9	P810	P912
N8	P811	P913
P8	P812	P914
B1	P813	P915

U1G
R7KA8P1KFLCAC#UC0

GPIO PA / PB		
G1	PA00	PB00
H4	PA01	PB01
F1	PA02	PB02
G2	PA03	PB03
D1	PA04	PB04
G3	PA05	PB05
C1	PA06	PB06
G4	PA07	PB07
F3	PA08	
F4	PA09	
F2	PA10	
B4	PA11	
B2	PA12	
C3	PA13	
D4	PA14	
E1	PA15	

U1H
R7KA8P1KFLCAC#UC0

GPIO PC / PD		
M1	PC00	PD00
M3	PC01	PD01
L2	PC02	PD02
L1	PC03	PD03
L3	PC04	PD04
L5	PC05	PD05
N4	PC06	PD06
K5	PC07	PD07
M4	PC08	
L4	PC09	
M5	PC10	
H5	PC11	
G5	PC12	
J5	PC13	
F5	PC14	
K1	PC15	

U1M
R7KA8P1KFLCAC#UC0

USB / MIPI		
K15	AVCC__USBH_S	AVCC_MIPi
U16	P814/U__SB_DP	MIPI_CL_N
T16	P815/U__SB_D_M	MIPI_CL_P
H17	USBH_S_DM	MIPI_DL0_N
H16	USBH_S_DP	MIPI_DL0_P
J15	USBH_S_RRE_F	MIPI_DL1_N
U17	VCC__U__SB	MIPI_DL1_P
H15	VCC__U__SBHS	VCC18__MIPi
T17	VSS__U__SB	VSS__MIPi
J17	VSS1__USBH_S	
J16	VSS2__USBH_S	

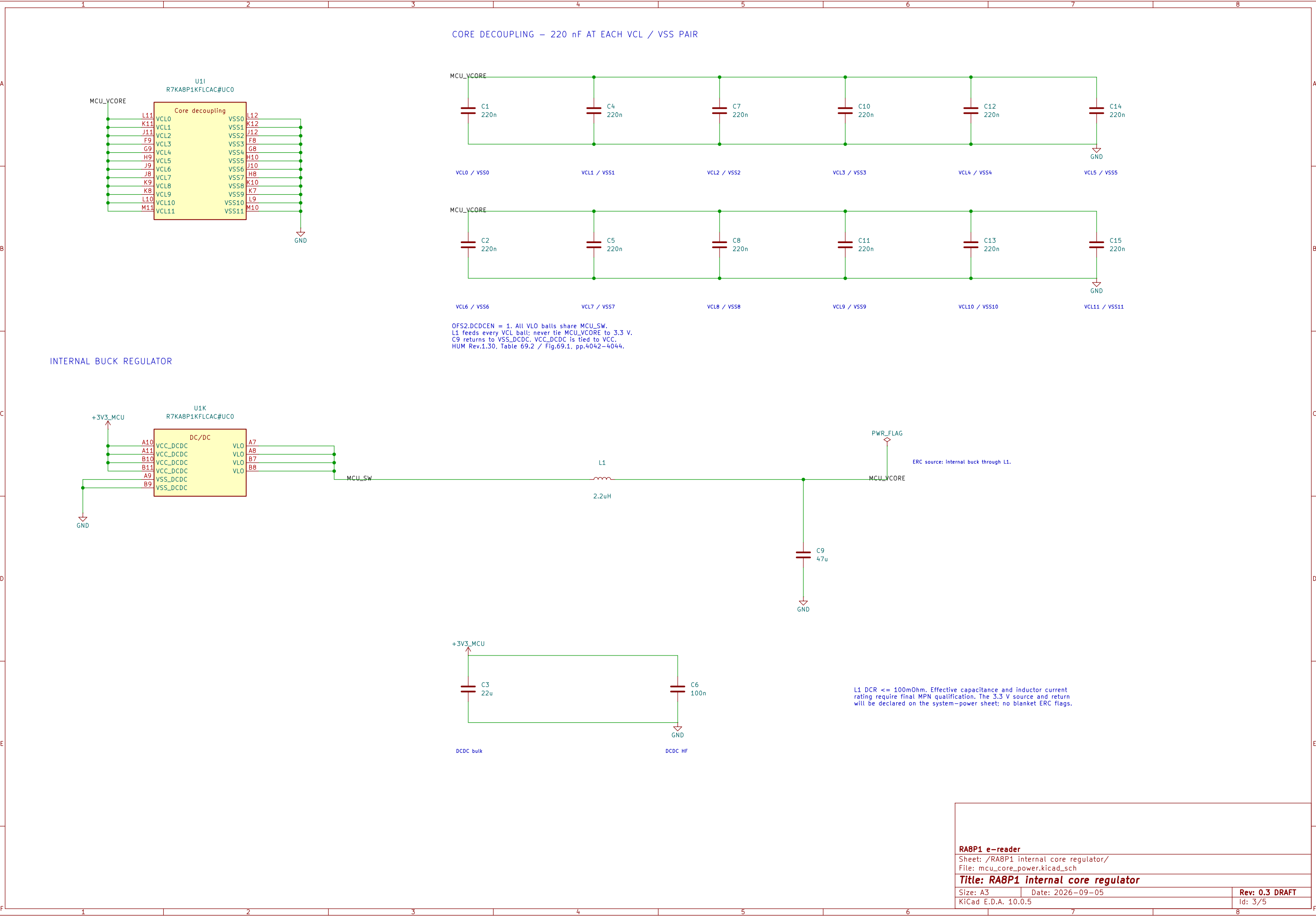
RA8P1 e--reader

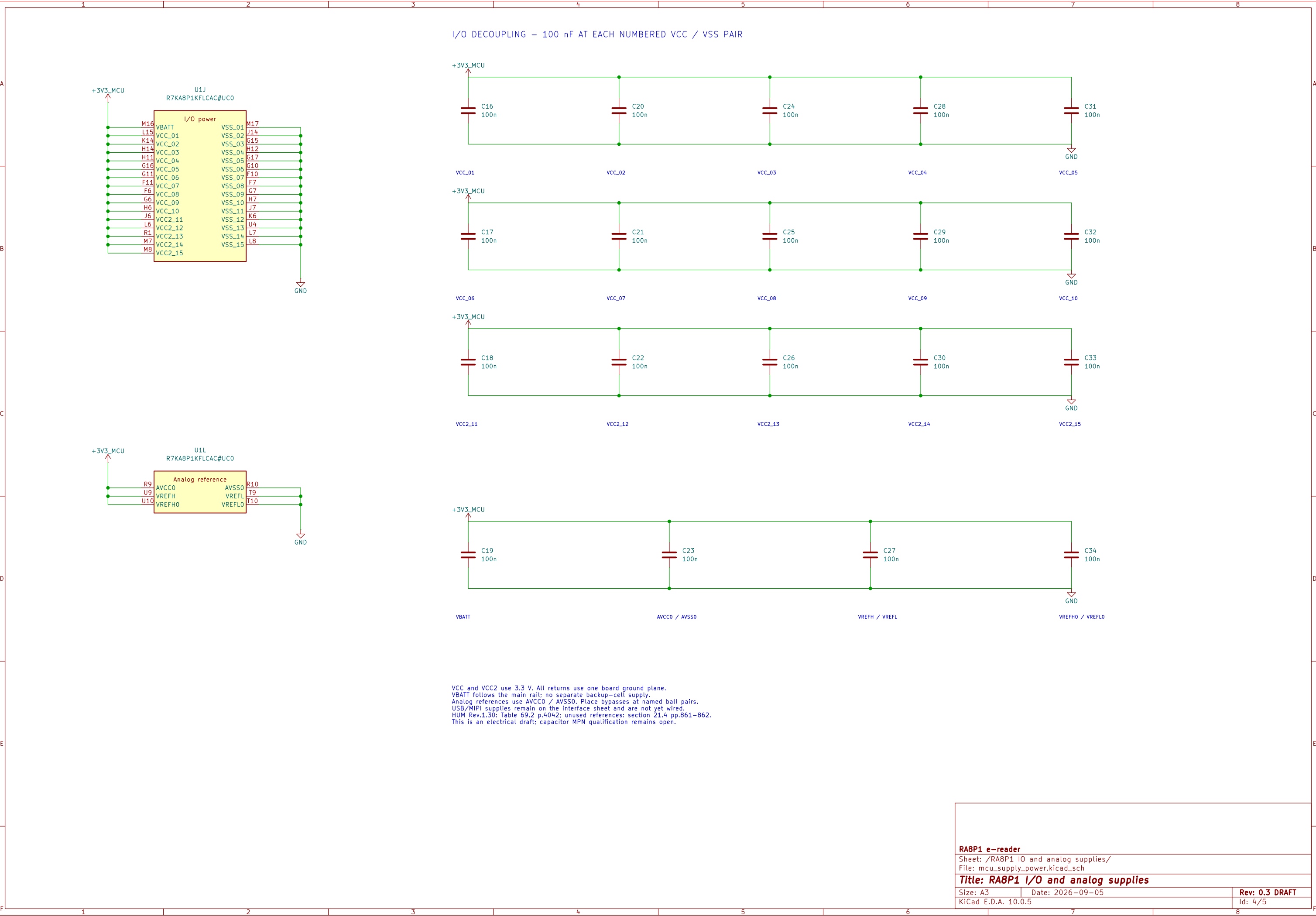
Sheet: /RA8P1 IO allocation/
File: mcu_interfaces.kicad_sch

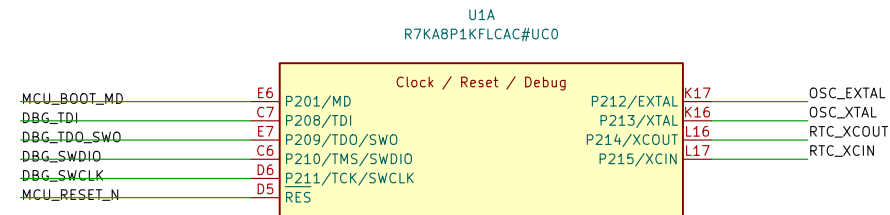
Title: RA8P1 IO allocation

Size: A3 Date: 2026-09-05
KiCad E.D.A. 10.0.5

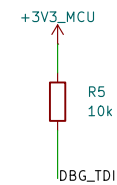
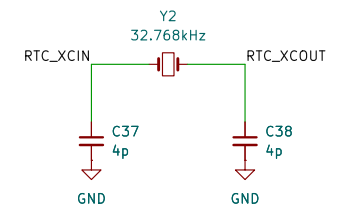
Rev: 0.3 DRAFT
Id: 2/5







```
CLK-002: Y2 ABS07-LR-32.768KHZ-6-1-T
Crystal CL=6 pf; Cstray=6 pf ASUMED.
CL=C37*C38/(C37+C38)+Cstray
C37=C38*2*(6-4)=4 pf, +/-0.25 pf NPO.
Check: 4*4/(4+4)+4=6 pf.
Fixed-stray tolerance: 5.875-6.125 pf.
Initial mode: SODRV=10b; SOSEL=0.
Qualify startup, frequency, drive <=0.5 uW.
Full math/Python: ../design/clock_calculations.md
```



References: RA8P1 HUM Rev.1.30 sections 2.5, 4.3–4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RA8P1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.